

RINGS, FIELDS AND GALOIS THEORY – A PROTOTYPE

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This text contains the topics covered in the lectures of Honours Algebra IV (Math 457) at McGill university in the winter semester of 2018. The instructor of this course is Professor Marcin Sabok. These notes are rather terse and will be replaced by a complete course text near the end of the winter semester.

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Course Outline and Syllabus. Since the course outline was only printed and handed students (i.e. not posted on the course page) I believe it is wise to reiterate some of the important parts here. This only concerns students enrolled in the course mentioned above. For the most part, we cover the following topics

- (1) Rings, ideals and quotient rings. The Chinese remainder theorem, Euclidean and Principal Ideal Domains will also be studied.
- (2) Modules, submodules and free modules are central to this section. The corresponding Chinese remainder theorem and elementary divisors theorem are also of interest.
- (3) Fields, characteristics, extensions, degrees and applications to classical geometry. Algebraic extensions, splitting fields and algebraic closure are also studied.
- (4) Galois theory, automorphisms over subfields and Galois groups. Galois extensions and the fundamental theorem of Galois theory will be covered.

The course follows the Algebra IV lecture notes written by Professor Eyal Goren. Another recommended resource is Dummit and Foote. There will be weekly

homework assignments and a midterm examination on February 21st, 2018. The final grade is the maximum of the following two options

15% homework, 25% midterm and 60% final exam;

15% homework and 85% final exam.

An advanced undergraduate background in the linear algebra and group theory is assumed throughout this text.

Part 1. Ring Theory

Throughout this part of the text, we shall denote by $\mathcal{R}$ a non-empty set. In practice, it will be used to describe a *ring*, which we define below.

1. RINGS AND IDEALS

DEFINITION 1. Let $\mathcal{R}$ be a non-empty set and suppose we are given two binary operations $+, \cdot$ from $G \times G \longrightarrow G$. We say the tuple $(\mathcal{R}, +, \odot)$ is a **ring** provided $(\mathcal{R}, +)$ is an Abelian group, $(\mathcal{R}, \cdot)$ is a monoid, and

- (1) $x \cdot (y + z) = x \cdot y + x \cdot z$,
- (2) $(x + y) \cdot z = x \cdot z + x \cdot y$

for all $x, y, z \in \mathcal{R}$. We denote by 0 and 1 the identity elements of $(\mathcal{R}, +)$ and $(\mathcal{R}, \cdot)$, respectively.

For the sake of simplicity, we will write xy to denote $x \cdot y$, for elements of a ring $\mathcal{R}$. It is useful to present some examples of rings that the reader should be familiar with. The $+$ operation is referred to as “addition” and $\cdot$ is deemed a form of “multiplication”. Finally, we omit the tuple notation and will write $\mathcal{R}$ to mean $(\mathcal{R}, +, \cdot)$.

- If $\mathbb{F}$ is a field, then it is a ring. This includes $\mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ as special cases.
- The set of integers, $\mathbb{Z}$, is a ring under the obvious notions of addition and multiplication.
- For every $n \in \mathbb{Z}$, the set $\mathbb{Z}/n\mathbb{Z}$ is a ring in the obvious way. Notice that if n is not prime, then $\mathbb{Z}/n\mathbb{Z}$ is not a field!
- Let $\mathbb{F}$ be a field and V a vector space over $\mathbb{F}$. The set $\text{End}(V)$ is a ring where we define for $f, g \in \text{End}(V)$

$$(f + g)(v) := f(v) + g(v), \quad (fg)(v) := (f \circ g)(v)$$

for all $v \in V$.

- If $\mathcal{R}$ is a singleton then it can be given a ring structure. Denote the element of $\mathcal{R}$ by 0 and define $0 + 0 = 0$ and $0 \cdot 0 = 0$. Then, this becomes a ring

whose multiplicative identity is 0. This ring is called the **zero-ring** and is clearly unique (up to a relabeling of the element).

There is yet more terminology to introduce.

DEFINITION 2. If a set $\mathcal{R}$ is a ring, then we say

- $\mathcal{R}$ is a commutative ring if $xy = yx$ for all $x, y \in \mathcal{R}$;
- The set of units in $\mathcal{R}$, is the non-empty¹ collection of elements having a multiplicative inverse:

$$\mathcal{R}^\times := \{x \in \mathcal{R} : \exists y \in \mathcal{R} \text{ such that } xy = yx = 1\};$$

- An element $x \in \mathcal{R}$ is called a zero-divisor if there exists $y \in \mathcal{R}$ such that $xy = 0$ or $yx = 0$;
- A non-zero commutative ring is called an integral domain if there are no zero-divisors;
- A non-zero ring is called a division ring if $\mathcal{R}^\times = \mathcal{R} \setminus \{0\}$. A field is therefore a commutative division ring.

We now summarize some immediate consequences of the ring axioms.

- (1) The respective identities 0 and 1 are unique. First, 0 is unique as the identity of a group is unique. For 1, suppose $1' \in \mathcal{R}$ also satisfies $1' \cdot x = x \cdot 1' = x$ for all $x \in \mathcal{R}$. Then,

$$1 = 1 \cdot 1' = 1'.$$

- (2) $0 \cdot x = x \cdot 0 = 0$, for all $x \in \mathcal{R}$. Indeed,

$$0 \cdot x = (0 + 0) \cdot x = 0 \cdot x + 0 \cdot x$$

and

$$x \cdot 0 = x \cdot (0 + 0) = x \cdot 0 + x \cdot 0$$

from which the result follows.

- (3) If $\mathcal{R}$ is a ring such that $0 = 1$ (as formal symbols), then $\mathcal{R} = \{0\}$. To see this, let $x \in \mathcal{R}$ and write $x = 1 \cdot x = 0 \cdot x = 0$.

1.1. Ideals. Let $\mathcal{R}$ be a ring and I a subgroup of $(\mathcal{R}, +)$. We say that I is an ideal in $\mathcal{R}$ (or a two-sided ideal) provided for all $x \in \mathcal{R}$ and $r \in I$ there holds

$$xr \in I \quad \text{and} \quad rx \in I.$$

In any ring $\mathcal{R}$, the sets $\{0\}$ and $\mathcal{R}$ are themselves ideals which we henceforth deem the trivial ideals. We will then write $I \triangleleft \mathcal{R}$. We point out that this notation is no mistake. Since I is a subgroup of $(\mathcal{R}, +)$, it is clear that $\mathcal{R}$ is non-empty. Also, since $\mathcal{R}$ is commutative with respect to addition, I is a normal subgroup of $\mathcal{R}$.

¹Notice that $1 \cdot 1 = 1$.

EXAMPLE 1.1. Let $\mathcal{R}$ be a commutative ring and fix $r \in \mathcal{R}$. The principal ideal generated by r is

$$(r) := \{rx : x \in \mathcal{R}\} \triangleleft \mathcal{R}.$$

We now prove that this is an ideal. If rx and ry live in (r) , then $rx + ry = r(x + y)$ also belongs to (r) . Also, if $a \in \mathcal{R}$ then

$$a(rx) = r(ax) \in \mathcal{R}.$$

Our first result is reminiscent of group theory and concerns quotient rings.

PROPOSITION 1.1. *Let $\mathcal{R}$ be a ring and I an ideal in $\mathcal{R}$. The quotient group $\mathcal{R}/I$ may be given a ring structure by defining*

$$(x + I) \cdot (y + I) := xy + I \tag{1.1}$$

for elements of $\mathcal{R}/I$.

Proof. We know from the theory of groups that the quotient $\mathcal{R}/I$ is well defined. We first check that the definition in (1.1) is well defined. Suppose that $x, x', y, y' \in \mathcal{R}$ are such that

$$(x + I) = (x' + I) \quad \text{and} \quad (y + I) = (y' + I).$$

Then, $x' = x + i_1$ and $y' = y + i_2$ for some $i_1, i_2 \in I$ so that

$$x'y' = (x + i_1)(y + i_2) = xy + i_1y + xi_2 + i_1i_2.$$

Since $I \triangleleft \mathcal{R}$, the sum $i_1y + xi_2 + i_1i_2$ belongs to I . Therefore, since I is normal in $(\mathcal{R}, +)$, we find that

$$(x'y' + I) = (xy + i_1y + xi_2 + i_1i_2 + I) = xy + I.$$

This shows that the multiplication operation is well defined. Now, it is clear from the ring structure of $\mathcal{R}$ that $\mathcal{R}/I$ is a ring under this notion of multiplication. $\square$

Although the above proposition gives us a notion of “normality” in rings, it is not useful in studying the structure of ideals.

PROPOSITION 1.2. *Suppose that $\mathcal{R}$ is a ring and I an ideal in $\mathcal{R}$. Then $0 \in I$ and $I = \mathcal{R}$ if $1 \in I$. More generally, if I contains an element of $\mathcal{R}^\times$ then $I = \mathcal{R}$.*

Proof. The claim is clear if $\mathcal{R}$ is the zero-ring. Otherwise, note that I always contains 0 as $0 = x \cdot 0$. Suppose that $I \cap \mathcal{R}^\times$ is non-empty and choose x from their intersection. There exists $y \in \mathcal{R}$ such that $xy = yx = 1$. Since $\mathcal{R}$ is not the zero-ring, $y \neq 0$. Then, $xy = 1 \in I$ which means that $r = 1 \cdot r \in I$ for all $r \in \mathcal{R}$. $\square$

The following fact relates finite integral domain to fields.

PROPOSITION 1.3. *Let $\mathcal{R}$ be a finite integral domain, then $\mathcal{R}$ is a field.*

Proof. If we look back, we see that integral domains exclude the zero ring. Thus, $\mathcal{R}$ contains at least two elements, and thus $1 \neq 0$. Since $\mathcal{R}$ is commutative by default, we need only check that every element $x \in \mathcal{R}$ admits a multiplicative inverse. To this end, we fix $x \in \mathcal{R}$ and consider the map

$$\Gamma : \mathcal{R} \longrightarrow \mathcal{R}, \quad r \mapsto xr.$$

Then Γ is an injection since $xr_1 = xr_2$ implies $x(r_1 - r_2) = 0$ whence it follows that $r_1 = r_2$ (after all, $\mathcal{R}$ is an integral domain). As an injection between finite sets of the same cardinality, there exists $x^{-1} \in \mathcal{R}$ such that $\Gamma(x^{-1}) = 1$. $\square$

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